

Digital Brolly Voice Agent

Operator Guide & Build Report

Prepared 1 September 2026 · Telugu / Tenglish outbound calling · Sarvam voice backend

What this system does

It reads leads from your Google Sheet, calls each one in Telugu (or Tenglish, Hindi, English), either **delivers a message and hangs up (one-way)** or **holds a conversation about the courses (two-way)**, and writes the transcript and outcome back to the Sheet. Every call opens with an AI disclosure, respects DND, and uses the registered 140-series caller ID.

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Part 1 — Operator guide

1.1 Starting the system

Everything runs in Docker. From the project folder:

```
docker compose up -d --build
```

This starts five services:

Service	What it is	Where
backend	The calling engine (FastAPI). Dials, talks, records, syncs the Sheet.	<code>http://localhost:8091</code>
console	The web console you operate from.	<code>http://localhost:3200</code>
cloudflared	The public tunnel Plivo uses to reach the backend during a call.	automatic
redis	The call queue. Safe to flush; everything rebuilds from the database.	internal
postgres	The database — the single source of truth for leads, campaigns and calls.	internal

After changing code, always rebuild

`docker compose restart` and `--force-recreate` silently keep running the *old* image. Only `docker compose up -d --build` picks up code changes. Changes to `.env` do apply on a plain restart.

If calls ring and drop the moment they are answered — the tunnel died

Cloudflare quick tunnels get torn down without warning; `cloudflared` stays "Up" serving a dead hostname, and Readiness shows "*PUBLIC_BASE_URL ... is unreachable (ConnectError)*" for every campaign at once. Fix:

```
docker compose restart cloudflared
```

That is all — the backend asks the tunnel for its new hostname by itself before every dial batch (happened 1 Sep, 19:35 IST; recovered this way). The permanent fix is a **named** Cloudflare tunnel on your own domain (fixed hostname, needs a Cloudflare account): set `TUNNEL_TOKEN` and `PUBLIC_BASE_URL` in `.env`.

Open the console and check the **Readiness** strip at the top of the home page. Every item must be green before you dial. It checks the Plivo number, the tunnel, the Sarvam key, the database, Redis, and — for two-way Telugu — that the feature flag is on.

1.2 Preparing the Google Sheet

The Sheet is connected through the Apps Script already deployed against it (the `/exec` URL in `.env`). The backend reads leads from a tab named `Leads` every 10 minutes, and writes transcripts to a tab named

Transcripts .

The Leads tab did not exist — this is why no lead was ever called from the Sheet

Until 1 September the Sheet had no `Leads` tab. The script answered a missing tab exactly like an empty one, so the backend reported "synced 0, no error" every ten minutes for weeks. Create the tab as below, then apply the one-line script patch in section 2.2 so this can never be silent again.

Create a worksheet named exactly `Leads` . Row 1 is the header:

Name	Phone	Campaign	Language	Consent Basis	Consent At	Status	Notes
Mouli	7993399336	Demo Class Outreach	te	explicit	2026-09-01		

Column	Required	Rules
Phone	yes	Any Indian format. Normalised to +91. A row without a valid number is skipped.
Campaign	yes	Must name a campaign that already exists in the console. Capitals and spaces don't matter. Unknown name → row skipped.
Consent Basis	to dial	<code>explicit</code> or <code>inferred</code> . Anything else, and the lead imports but is never called .
Consent At	to dial	A date such as <code>2026-09-01</code> . A future date is rejected.
Name	no	Spoken in the greeting. English-letter names are converted to Telugu script so the voice pronounces them correctly (Mouli → మౌలీ).
Language	no	<code>te</code> , <code>tinglish</code> , <code>en</code> , <code>hi</code> , <code>hinglish</code> , or blank.
Status / Notes	no	Written by the agent after each call. Create the columns if you want that.

Consent decides whether a lead is dialled

Both **Consent Basis** and **Consent At** must be present and valid. A sheet without them imports every row and calls nobody. The console's Settings page now says this in plain words: *"No imported lead can be dialled — check the Consent columns."*

Header names are matched loosely (`Mobile` , `Contact` , `Phone Number` all work for the phone). Only row 1 is read as the header.

1.3 Creating a campaign

Console → **Campaigns** → **New campaign**. Three choices matter, and none of them can be changed by the AI during a call:

Setting	Options	What it decides
Language	te Telugu · tinglish Telugu+English · hi · hinglish · en · auto	Which voice engine runs. Telugu and Tenglish use Sarvam (Indian voices, Indian telephony STT). Hindi/English/Hinglish use ElevenLabs. ElevenLabs cannot speak Telugu at all — this is not configurable.
Mode	oneway · twoway	One-way: speaks the script and hangs up; the lead's voice is never listened to. Two-way: a real conversation — the lead asks about fees, duration, batches, and the agent answers from your course documents. When unsure, choose two-way.
Script	English text	Required for one-way. Written in English; the system renders it into natural Telugu once per campaign. For two-way it is optional — leave it blank and the agent opens with a standard greeting in the campaign's register (Telugu, or Tenglish with <i>course fees, batch timings, placement</i> in English) and takes questions.

The AI disclosure is enforced automatically. If a script lacks one, the system prepends "ఇది డిజిటల్ బ్రోలి నుండి ఏవ ద్వారా చేసే ఆటోమేటెడ్ కాల్" and logs a warning; after every call it re-checks what was actually spoken.

1.4 Loading leads

Three ways, all ending in the same place:

Method	How	Best for
Google Sheet	Fill the Leads tab. Sync runs every 10 minutes, or press Sync now on the Settings page.	Your normal daily flow.
CSV upload	Campaigns → open a campaign → Upload leads . Same columns as the Sheet.	A one-off list from another source.
Single lead	Campaign page → Add lead .	A test call to your own number.

After a Sheet sync, read the two numbers on the Settings page: **imported** (rows written) and **dialable** (rows the dialer will actually call). If dialable is 0 and imported is not, the consent columns are the reason.

1.5 Launching and watching calls

1. Campaign is **Active** (toggle on the campaign page).
2. Leads are **pending** with valid consent.
3. The scheduler dials automatically every minute during calling hours. Press **Dial now** on the campaign page to start immediately.

Watch progress on **Calls**. Each row shows the lead, status, duration and turn count; click through for the full transcript.

Lead status	Meaning
pending	Waiting to be dialled.
queued / calling	Picked up by the dialer / on the phone right now.
done	Call completed. Transcript stored.
failed	Not connected (no answer, busy, network). Retried up to max attempts (default 2).
dnd	On the Do-Not-Disturb registry. Never dialled.

Calling hours are set fully open (00:00–24:00) on the owner's instruction

India's TCCCPR generally restricts promotional calls to about 09:00–21:00, and penalties attach to the Plivo number and its DLT registration — a blacklisted number stops every campaign at once. Restoring a 10:00–19:00 window is one line in `.env` (`CALLING_HOURS_START` / `CALLING_HOURS_END`) and a restart.

1.6 Reading results

- **Console** → **Calls**: the authoritative record. Transcript, duration, turns, outcome, and the **Summary** — an outcome tag plus what the lead said, e.g. *"Interested — asked the BDCP fee, wants batch timings."*
- **Sheet** → **Transcripts tab**: a copy of every transcript, written within 30 minutes of the call. On the Leads tab, `Status` becomes the call result and `Notes` becomes that same summary, for the matching phone number.
- **Test on a handset, not a speakerphone in a room**. Background voices and echo trigger the recogniser; the noise gate now filters most of it, but a quiet handset is what a real lead sounds like.
- **Test-call checklist** for a two-way Telugu call to your own number:
 1. Say "ఓకే" mid-conversation — the call must **continue**.
 2. Ask a fee question — the agent must answer from the documents, not promise a callback.
 3. Say "ఓకే" — the call must end **with a spoken goodbye**.

1.7 Stopping

To...	Do
Stop one campaign	Toggle it Inactive . Calls in progress finish; nothing new is dialled.
Stop everything now	<code>docker compose stop backend</code> . Leads mid-call are released back to <code>pending</code> automatically.
Never call a number again	Leads → the lead → Do not call .

1.8 Daily checklist

1. Readiness strip all green.
2. Settings → Sheets sync: last sync recent, **no error**, dialable > 0.
3. Campaign active, script correct, mode correct.
4. One test call to your own number before real leads.
5. Check the log for `[compliance]` — any ERROR there is a **stop-dialling** event.

1.9 Five-minute demo

Two campaigns are ready and **inactive**, each with one lead (your own number, consent recorded): **Demo - Telugu two-way** and **Demo - Tinglesh two-way**. Nothing dials until you switch one on.

1. Home page: Readiness strip all green.
2. Campaigns → tick **Include inactive** (inactive campaigns are hidden by default) → **Demo - Telugu two-way** → toggle **Active** → **Dial now**. Your phone rings within seconds.
3. Answer on a handset. Say "ఓకే" (the call continues), ask "కోర్సు ఫీజు ఎంత?" (answered from the documents), say "బై" (farewell, then hang-up).
4. Calls page: the row appears with duration, turns and the **Summary** line. Click through for the transcript.
5. Toggle the campaign **Inactive** again. Repeat with the Tinglesh one to hear the difference: *course fees, batch timings, placement* in English.
6. Settings: Sheets sync status — shows what the Sheet integration reports (needs the **Leads** tab, \$1.2).

Part 2 — Build report, 1 September 2026

2.1 What was found broken

#	Problem	Evidence
1	The Sheet fed no leads	<code>/admin/sheets-status</code> reported <code>synced 0, error null</code> every 10 minutes. The container log held the real cause: <code>update_lead error: 'Leads sheet not found.'</code> The spreadsheet had no <code>Leads</code> tab, and the script reported a missing tab exactly like an empty one.
2	The last word of sentences was deleted	A guard that strips unpronounceable scripts (Armenian, Hindi) judged whole words by every character. The Telugu sentence mark <code>l</code> was not on its allowed list, so <code>ఉన్నాయిl</code> became nothing. The codebase's own sentence splitter treats <code>l</code> as normal — the two rules contradicted each other. This was the "words breaking in the middle" you heard.
3	Calls hung up on "okay"	The model called <code>end_call</code> when the lead said <code>"అ ఓకే"</code> — agreement, not goodbye.
4	One-way calls mispronounced names	Transcripts show <code>నమస్తే Mouli గారు</code> — the English spelling reached the Telugu voice. The fix existed only on the two-way path.
5	Consent column collision	A sheet with only a "Consent At" column matched it as the consent <i>basis</i> too, filing a date where "explicit" belongs — leads permanently undialable, nothing logged. The CSV importer already guarded this; the Sheet path did not.
6	Sync reported the wrong number	It counted rows written, not rows the dialer would call. With no consent columns those are "40" and "0" — and it showed 40.
7	No manual sync	The only trigger was the 10-minute timer. After fixing a sheet, you waited to find out.
8	Campaign names matched exactly	"demo class" in the Sheet did not match "Demo Class" in the console — row silently skipped.
9	Sheet leads dialled last	They carried no dial order, so every uploaded lead was called before any Sheet lead.
10	Tinglish contradicted itself (fixed, rows 14–16)	Tinglish differs from Telugu by one instruction ("keep course, fees, batch... in English"), but three later rules order the model to reply in Telugu script and to spell <i>course</i> , <i>fees</i> , <i>placement</i> in Telugu. Tinglish two-way has never been run: 2 campaigns, 1 lead, one-way only.

2.2 What was changed

#	Change	Where
1	Missing Leads tab is now an error , not a healthy zero	Backend: <code>app/sheets/apps_script.py</code> raises on it. Script side needs your one-line patch — see box below.
2	Script guard judges letters only ; punctuation never disqualifies a word	<code>app/telephony/sarvam_prompts.py</code> <code>_is_speakable()</code>
3	<code>end_call</code> refused when the lead's last words are pure agreement (ఓకే / సరే / అవును); real goodbyes — including STT's "బాయ్" spelling — still end the call with a farewell. Capped at 2 refusals.	<code>app/telephony/sarvam_bridge.py</code> <code>_AFFIRMATION_ONLY</code> , <code>_lead_only_agreed()</code>
4	One-way greeting resolves the name to Telugu script via the same <code>telugu_name()</code> the two-way path uses	<code>sarvam_bridge.py</code> <code>_spoken_name()</code>
5	Consent-column collision guard, copied from the CSV importer	<code>app/sheets/sync.py</code>
6	Sync returns <code>received</code> / <code>imported</code> / <code>dialable</code> / <code>skips</code> ; the console shows all four and warns when <code>dialable</code> is 0	<code>sync.py</code> <code>SyncResult</code> , <code>app/admin/system.py</code> , <code>frontend/app/settings/page.tsx</code>
7	Sync now button and <code>POST /admin/sheets/sync</code>	<code>app/admin/system.py</code> , settings page
8	Campaign name matched ignoring case and spaces	<code>app/db/campaigns.py</code>
9	Sheet row number becomes the dial order	<code>sync.py</code>
10	Consent basis lowercased on import ("Explicit" now works)	<code>sync.py</code>
11	<code>/admin/sheets-status</code> answers "can the sync run" instead of "is the env var non-empty"	<code>app/admin/system.py</code>
12	Console added to <code>docker compose</code> (port configurable via <code>CONSOLE_PORT</code>)	<code>docker-compose.yml</code> , <code>frontend/Dockerfile</code>
13	Voice speed raised 1.0 → 1.2 (about 25% shorter agent turns)	<code>.env</code> <code>SARVAM_TTS_PACE</code>
14	Tinglish prompt no longer contradicts itself. The OUTPUT FORMAT, TELUGU SPELLING and UNCLEAR SPEECH rules and both	<code>app/languages.py</code> , <code>app/telephony/sarvam_prompts.py</code>

	script framings are now chosen by register: for Tenglish they no longer demand "Telugu script" or mandate Telugu spellings for <i>course</i>, <i>fees</i>, <i>placement</i>. Pure Telugu is unchanged. The register is read back off the style text with <code>languages.is_code_mixed()</code> — exact match against the catalogue, so a placeholder or a missing style is always pure Telugu.	
15	A Tenglish opening. Scriptless two-way calls had one Telugu constant for both registers. <code>DEFAULT_TWOWAY_TINGLISH</code> keeps the identical disclosure sentence (అటోమేటెడ్ కాల్, never a Latin "AI") and then "మా digital marketing courses గురించి... Course fees, batch timings, placement...". Chosen per call by <code>default_twoway_opening()</code>.	<code>sarvam_prompts.py</code> , <code>sarvam_bridge.py</code>
17	Noise gate on the lead's audio. Sarvam's "speech started" signal fires on <i>any</i> sound and was the agent's only interrupt signal, so a cough, a TV or a second person cut the agent off mid-sentence exactly like the lead's voice, and the recogniser even produced fake sentences from room noise. A "speech started" now counts only after 200 ms of audio at $\geq 1.5\times$ the quiet level measured just before it (never below an absolute minimum); a transcript from a sound the gate <i>measured</i> as non-speech is dropped, but anything it could not measure passes through — ignoring a real lead is the worse failure. Every decision is logged as <code>noise gate:</code> with its numbers; all thresholds are <code>.env</code> knobs; <code>SARVAM_NOISE_GATE_ENABLED=false</code> restores the old behaviour. Real interruptions now land ~200 ms later. The floor is the <i>quietest</i> 200 ms of recent non-speech audio, not the average — on the rehearsal call the average was inflated by the agent's own echo on the handset and a real question was dropped once.	<code>app/telephony/sarvam_stt.py</code> , <code>app/config.py</code>

18	A summary for every call. Sarvam calls never had one — the console's Summary column was blank and the Sheet's Notes said "N turns". After each call the model now writes a two-line English note from the transcript: an outcome tag (<i>Interested / Not interested / Callback / No answer / Unclear</i>) and what the lead asked or said, e.g. "Callback — asked the BDCP fee (₹50,000), wants a call tomorrow." Written after the call outcome, never delaying it; a one-way call gets "One-way message delivered."; it uses only the transcript and is told never to invent a fact or price.	<pre>app/telephony/call_summary.py , call_routes.py , sheets/writeback.py</pre>
16	Tests on the real catalogue text. The Sarvam prompt tests had used hand-written placeholder styles, which is why the contradiction survived. 15 new tests use <code>languages.style("tinglish") / ("te")</code> directly and assert each register's prompt is internally consistent.	<pre>tests/unit/test_sarvam_prompts.py , test_sarvam_bridge.py , test_languages.py</pre>

Three deliberate non-changes in the Tinglesh work

- **Speech recognition stays pinned to `te-IN`** for both registers. Sarvam auto-detection mis-hears Telugu as Croatian/Urdu; the register lives in the prompt, not the transport, and `te / tinglish` must share one backend switch (`_TELUGU_TOKENS`) or preflight breaks.
- **The short canned lines** — farewell, "say that again", "one moment", the disclosure prefix — stay in Telugu. A Hyderabad Tinglesh speaker says every one of them exactly so, and the disclosure prefix must stay in Telugu for compliance.
- **"SEO" is still spoken as ఎస్ఈఓ** in both registers: an acronym is pronounced by its letter names in any language, and Latin "SEO" would be read by the synthesiser as a word.

One action for you: patch the Apps Script (2 minutes)

In the Sheet: **Extensions** → **Apps Script**, find `_getLeads`, and replace

```
if (!sheet) return _json({ status: "ok", leads: [] });
```

with

```
if (!sheet) return _json({ status: "error", message: "Leads sheet not found: " + name });
```

Then **Deploy** → **Manage deployments** →  → **Deploy**. The URL does not change. Full instructions: [docs/sheet-setup.md](#).

2.3 How it was verified

- **1,224 unit tests and 40 integration tests pass** (the integration suite ran against the separate `voiceagent_test` database), `ruff` clean. Every change above has a test that fails without it.
- Danda bug: proven with the exact deployed regex — ఉన్నాయి, ఉంటుంది, ఫీజు... all DROPPED before, all kept after; Armenian and Hindi still dropped.
- Manual sync exercised live against the backend: `{"synced":0,"received":0,"dialable":0,"error":null}` — correct until the tab exists.
- Console container exercised live: serves the page and proxies to the backend with auth through the compose network.
- **Rehearsal call, 1 Sep 20:23 IST** (19 turns, 2 min 55 s, handset): fee, course list, topics, offline option and the fee breakdown all answered from the documents; "batch details" honestly declined (not in the documents); clean ending on "ఒకే థాంక్స్". The summary was written live: *"Interested — Lead asked about the digital marketing course topics, batch details, offline training, and what the ₹50,000 includes..."* One gate mistake found and fixed the same hour: the noise floor was being measured as an average that the agent's own echo inflated (see row 17); it is now the quietest 200 ms of recent non-speech audio.
- **Tinglish two-way proven on a live call, 1 Sep 17:01 IST** (9 turns, 49 s): the opening was spoken exactly as designed — ...మా digital marketing courses గురించి కాల్ చేస్తున్నాను. Course fees, batch timings, placement, ఏమైనా తెలుసుకోవాలంటే అడగండి — and the reply to "what courses are there?" came back as natural Hyderabad Tinglish: డిజిటల్ మార్కెటింగ్ లో BDLP, BDCP, BDDP, BDMP courses ఉన్నాయి. మీరు ఏ course గురించి details కావాలనుకుంటున్నారు? Disclosure first, name correct, no compliance warning.
- **Telugu two-way proven on a live call, 1 Sep 16:21 IST** (13 turns, 1 m 40 s, owner and team lead listening): name spoken as మౌలి గారు, disclosure first, an off-topic question (sports) refused politely, the **fee question answered directly from the documents** — యాభై వేల రూపాయలు, 25 k training + 25 k placement — and a clean `end_call` exit with a spoken farewell. Per-turn: LLM 1.0–2.1 s, TTS 0.24 s, total 1.4–2.4 s.
- Live two-way call today opened "నమస్తే మౌలి గారు" with the disclosure first, and hung up cleanly when it reached voicemail rather than talking into it for five billed minutes.
- Latency measured on real turns: STT $\approx$ 0.0 s, gate 0.12 s, **LLM 0.8–2.9 s**, TTS 0.24–0.34 s. The model is 74–89% of the wait.

2.4 What remains

Item	Plan
Noise gate — tune from real calls	Built (row 17) from the numbers of one call. On the next noisy test, search the log for <code>noise gate</code> : . If the agent ignores a soft-spoken lead, lower <code>SARVAM_NOISE_GATE_SNR</code> (1.5 → 1.2) or <code>SARVAM_NOISE_GATE_MIN_RMS</code> ; if room sound still interrupts, raise them. The audio-health lines now end with <code>gate=pass noise unmeasured</code> per utterance.
Latency	The LLM is the wait. Next steps in order of value: trim the system prompt and course digest, then cache answers to the handful of repeated questions (fees, duration, timings). Sentence-by-sentence streaming was built and removed twice — it breaks the one-turn-at-a-time invariant the barge-in handling depends on.
Document gaps	Demo classes, batch timings and eligibility are not in the three ingested documents, so the agent promises a callback. Add them and re-run <code>scripts/ingest_docs.py</code> . Also: the BDCP document says "3 Months. Includes a 4 Months Internship" — a data error in the source.
Second test number	Gayathri's number (+91 77299 74373) was dialled twice at 16:21 and 16:28 IST and rang out both times (<code>NO_ANSWER</code>); the lead is back to <code>pending</code> and has used its 2 attempts. If that was the team lead's phone, re-add the lead or raise the campaign's max attempts.

2.5 Risks you are carrying

Risk	Detail
Public Sheet endpoint	The Apps Script <code>/exec</code> URL accepts writes from anyone who has it — no token is set on either side. Before the Sheet holds a real lead list, set a <code>SHEETS_TOKEN</code> script property and the matching <code>.env</code> value.
Calling hours open 24h	Owner's decision, 22 July 2026. Regulatory exposure attaches to the Plivo number. One-line revert.
Nothing committed	All of today's changes are in the working tree, uncommitted, on branch <code>dev1</code> . Commit when you are satisfied with the live test.
ElevenLabs billing is past due	Readiness reports the ElevenLabs account as <code>past_due</code> : no English, Hindi or Hinglish call can run until the subscription is paid. Telugu and Tinglish run on Sarvam and are unaffected.
Test campaigns retired	All 16 test campaigns were deactivated on 1 Sep evening (leads and call history kept; visible under "Include inactive"). Only the two inactive demo campaigns remain listed, so nothing can dial by accident.
212 stale test campaigns	The database holds 212 <code>auto</code> -language campaigns from earlier testing. Harmless, but they clutter the console. Deactivate or delete when convenient.

Source of truth for all behaviour: the repository's `CLAUDE.md` , `docs/sheet-setup.md` , and the test suite under `tests/unit/` .